Energy calculations: worked solutions

Year 8 SEAL Science · Chapter 7 Energy · Kai's Off-Grid Year · Lessons 1 to 7 · 69 marks + 7 bonus

How the marks work. Each line marked [1] earns one mark. There are no half marks. The final answer needs its unit. Follow-through marks: if a part needs an answer from an earlier part, a wrong earlier answer used correctly still earns the later marks. Any correct method earns full marks.

Common slip. Under multi-step parts, this line names the mistake students make most often, so you can check your own working for it.

Answers at a glance

Part	Answer	Mk	Part	Answer	Mk
A1	1,225 J	2	**D2(a)**	Pump 1 9.6 kWh, Pump 2 6.6 kWh	2
A2(a)	0.16 kg	1	**D2(b)**	Pump 1 $16.80, Pump 2 $11.55	2
A2(b)	98 J	2	**D3(a)**	X $91.00, Y $63.70	2
A3(a)	3^2 is 9, not 6	1	**D3(b)**	30%	2
A3(b)	7.2 J	1	**D4(a)**	5 kWh	1
A4	3 m/s	3	**D4(b)**	4 h	2
B1	105,000 J	2	**E1(a)**	1,350,000 J	2
B2(a)	300 m	1	**E1(b)**	1,800,000 J	2
B2(b)	225,000 J	2	**E1(c)**	450,000 J	1
B3(a)	14 J	2	**E1(d)**	1,200,000 J	2
B3(b)	20 m	2	**E2(a)**	135,000 J	2
B3(c)	3.5 J	2	**E2(b)**	Friction	1
C1(a)	65%	2	**E2(c)**	33,750 J	1
C1(b)	770 J	1	**E2(d)**	Moving air carries heat away by convection	2
C2(a)	60 J	1	**E3(a)**	2.1 kWh per hour	2
C2(b)	60%	2	**E3(b)**	No: the loss falls by 30%, not half	2
C2(c)	Conduction	1	**E3(c)**	10.8 kWh	1
C3(a)	700 J	1	**E3(d)**	R4.0	1
C3(b)	Kinetic 700 J, thermal 600 J, sound 100 J	3	**X1(a)**	2,025 J	2
C4(a)	A 5,000 J, B 2,700 J	2	**X1(b)**	30%	1
C4(b)	A 50%, B 80%; Hoist A	2	**X1(c)**	Thermal energy (or sound)	1
D1(a)	2,800 Wh	1	**X2(a)**	15.3%	1
D1(b)	2.8 kWh	1	**X2(b)**	The panel (20.4% vs 17.1%)	2
D1(c)	$0.98	1			

X1 and X2 are the extension (bonus) questions.

Section A: Kinetic energy (10 marks)

A1 Red kangaroo **Easy** (2 marks)

Calculate the kangaroo's kinetic energy. [2 marks]

KE = ½mv² = ½ × 50 × 7²	**[1]**
7² = 49, so KE = ½ × 50 × 49 = 25 × 49	
KE = 1,225 J	**[1]**

Common slip: *175 J means the speed was not squared (½ × 50 × 7).*

A2 Fast bowler **Medium** (3 marks)

(a) Convert the mass of the ball to kilograms. [1 mark]

160 g ÷ 1,000 = 0.16 kg	**[1]**

(b) Calculate the ball's kinetic energy as it leaves the bowler's hand. [2 marks]

KE = ½ × 0.16 × 35²	**[1]**
35² = 1,225, so KE = 0.08 × 1,225	
KE = 98 J	**[1]**

Common slip: *98,000 J means the mass was left in grams. Convert to kilograms before you substitute.*

A3 Spot the mistake: lawn bowls **Medium** (2 marks)

Tama's working: Line 1: KE = ½mv² → Line 2: KE = ½ × 1.6 × 3² → Line 3: KE = 0.8 × 6 → Line 4: KE = 4.8 J

(a) State what Tama did wrong in line 3. [1 mark]

Tama wrote 3² = 6. That is 3 × 2, not 3 × 3. 3² = 9.	**[1]**

(b) Calculate the correct kinetic energy of the bowl. [1 mark]

KE = 0.8 × 9 = 7.2 J	**[1]**

A4 Toboggan **Hard** (3 marks)

Calculate how fast the toboggan is moving. [3 marks]

KE = ½mv², so 450 = ½ × 100 × v²	**[1]**
450 = 50 × v², so v² = 450 ÷ 50 = 9	**[1]**
v = √9 = 3 m/s	**[1]**

Common slip: *9 m/s means the square root was not taken. 2.1 m/s means the ½ was left out.*

Section B: Gravitational potential energy and energy chains (11 marks)

B1 Lookout walk **Easy** (2 marks)

Calculate the gravitational potential energy she gains. [2 marks]

GPE = mgh = 70 × 10 × 150	**[1]**
GPE = 105,000 J (105 kJ)	**[1]**

B2 Chairlift **Medium** (3 marks)

(a) Calculate the height the skier rises. [1 mark]

h = 1,750 − 1,450 = 300 m	**[1]**

(b) Calculate the gravitational potential energy the skier gains. [2 marks]

GPE = mgh = 75 × 10 × 300	**[1]**
GPE = 225,000 J (225 kJ)	**[1]**

Common slip: *1,312,500 J means 1,750 m was used for h. Use the height gained, not the height above sea level.*

B3 Stomp rocket (6 marks)

(a) Calculate the rocket's kinetic energy at launch. [2 marks · Medium]

Convert: 70 g = 0.07 kg	
KE = ½ × 0.07 × 20² = ½ × 0.07 × 400	**[1]**
KE = 14 J	**[1]**

Common slip: *14,000 J means the mass was left in grams.*

(b) Assume no energy is wasted, so all of the kinetic energy becomes GPE at the top. Calculate the greatest height the rocket could reach. [2 marks · Hard]

GPE at the top = KE at launch = 14 J, so 14 = 0.07 × 10 × h	**[1]**
14 = 0.7 × h, so h = 14 ÷ 0.7 = 20 m	**[1]**

Marking note: *Mark consequentially: use your KE from part (a).*

(c) The rocket really reaches a height of 15 m, where it stops for an instant. Calculate how much energy was wasted on the way up. [2 marks · Hard]

GPE at 15 m = 0.07 × 10 × 15 = 10.5 J	**[1]**
Wasted = 14 − 10.5 = 3.5 J	**[1]**

Common slip: *20 − 15 = 5 is a difference in height (m), not an amount of energy (J).*

Section C: Efficiency and Sankey diagrams (15 marks)

C1 Electric chainsaw **Easy** (3 marks)

(a) Calculate the efficiency of the chainsaw. [2 marks]

Efficiency = 1,430 ÷ 2,200 × 100%	**[1]**
= 0.65 × 100% = 65%	**[1]**

(b) Calculate the energy the chainsaw wastes each second. [1 mark]

Wasted = 2,200 − 1,430 = 770 J	**[1]**

C2 Angle grinder **Medium** (4 marks)

(a) Calculate the missing amount of sound energy. [1 mark]

Sound = 900 − 540 − 300 = 60 J (the outputs add up to the input)	**[1]**

(b) Calculate the efficiency of the grinder. [2 marks]

Efficiency = 540 ÷ 900 × 100%	**[1]**
= 0.6 × 100% = 60%	**[1]**

Common slip: *540 ÷ 360 = 150% divides by the wasted energy instead of the total energy in. Efficiency can never be more than 100%.*

(c) Heat spreads from the hot motor through the grinder's metal casing. Name this type of heat transfer. [1 mark]

Conduction	**[1]**

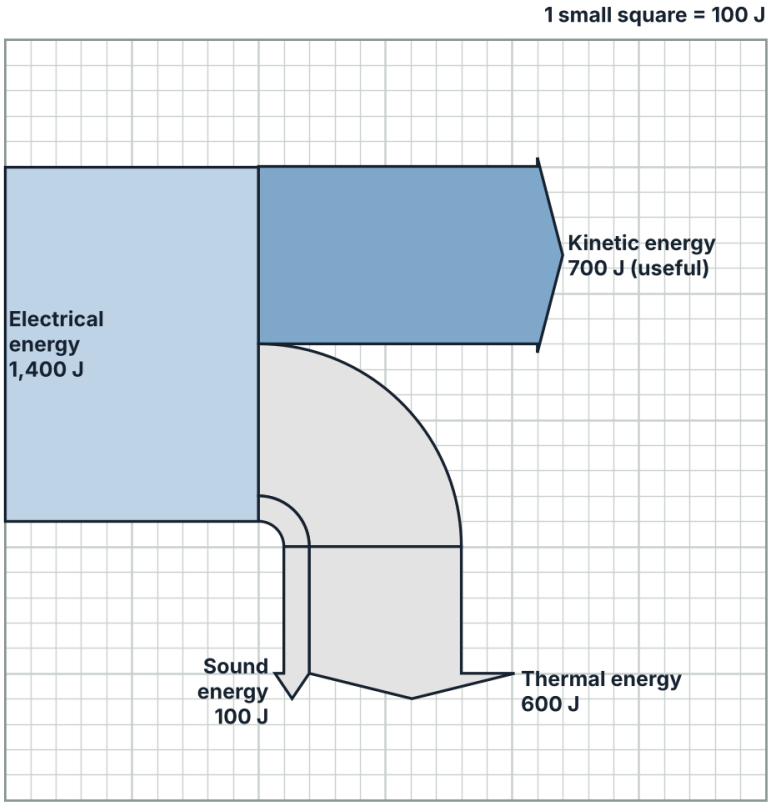
C3 Winch on a 4WD **Medium** (4 marks)

(a) Calculate the useful kinetic energy each second. [1 mark]

50% of 1,400 J = 0.50 × 1,400 = 700 J	**[1]**

(b) Work out the thermal and sound energy. Then complete the Sankey diagram on the grid. Start the useful arrow level with the top of the input arrow and draw it straight on to the right. Bend the wasted arrows down. Draw each arrow to scale and label it with its type and amount. A 1-square arrow can be a straight arrow pointing down, with its label beside it. [3 marks]

Wasted = 1,400 − 700 = 700 J. Split it into 7 equal parts (6 thermal + 1 sound): 700 ÷ 7 = 100 J. Thermal = 600 J, sound = 100 J.	**[1]**
Useful arrow: kinetic energy 700 J, 7 squares wide, straight on to the right, labelled.	**[1]**
Wasted arrows bend down: thermal 600 J (6 squares) and sound 100 J (1 square), labelled. All outputs add up to 14 squares, the width of the input.	**[1]**



Common slip: *If your output arrows add up to more than 14 squares, an amount is wrong. Thermal + sound must equal the 700 J wasted.*

C4 Two workshop hoists (4 marks)

(a) Calculate the useful energy for each lift: the GPE each load gains. [2 marks · Easy]

Hoist A: GPE = 250 × 10 × 2 = 5,000 J	**[1]**
Hoist B: GPE = 180 × 10 × 1.5 = 2,700 J	**[1]**

(b) Calculate the efficiency of each hoist. Then state which hoist wastes the larger share of its energy input. [2 marks · Hard]

Hoist A: 5,000 ÷ 10,000 × 100% = 50%. Hoist B: 2,700 ÷ 3,375 × 100% = 80%	**[1]**
Hoist A wastes the larger share: 50% of its input, against 20% for Hoist B	**[1]**

Marking note: *Mark consequentially: use your GPE values from part (a).*

Section D: Power, energy and running cost (14 marks)

D1 Boot dryer Easy (3 marks)

(a) Calculate the energy the rack uses overnight, in Wh. [1 mark]

Energy = 350 × 8 = 2,800 Wh	**[1]**

(b) Convert your answer to kWh. [1 mark]

2,800 ÷ 1,000 = 2.8 kWh	**[1]**

(c) Calculate the cost of running the rack for one night. [1 mark]

Cost = 2.8 × $0.35 = $0.98 (98 cents)	**[1]**

D2 Two pool pumps **Medium** (4 marks)

(a) Calculate the energy each pump uses per day, in kWh. [2 marks]

Pump 1: 1,600 × 6 = 9,600 Wh = 9.6 kWh	**[1]**
Pump 2: 550 × 12 = 6,600 Wh = 6.6 kWh	**[1]**

Common slip: *Comparing only the power (1,600 W and 550 W) ignores how long each pump runs.*

(b) Electricity costs 25 cents per kWh. Calculate the cost of running each pump for one week (7 days). [2 marks]

Pump 1: 9.6 × 7 = 67.2 kWh; 67.2 × $0.25 = $16.80	**[1]**
Pump 2: 6.6 × 7 = 46.2 kWh; 46.2 × $0.25 = $11.55	**[1]**

Marking note: *Mark consequentially: use your daily kWh from part (a).*

D3 Dehumidifiers (4 marks)

(a) Calculate the yearly running cost of each model. Use the table. [2 marks · Medium]

X: 350 × $0.26 = $91.00 per year	**[1]**
Y: 245 × $0.26 = $63.70 per year	**[1]**

(b) Calculate how much Model Y saves each year, as a percentage of Model X's running cost. [2 marks · Hard]

Saving = $91.00 − $63.70 = $27.30 per year	**[1]**
$27.30 ÷ $91.00 × 100% = 30%	**[1]**

Marking note: *Also correct: 245 ÷ 350 = 0.7, so Y uses 70% of X's energy and saves 30%.*

D4 Air compressor (3 marks)

(a) Calculate how many kWh of electrical energy $1 buys. [1 mark · Medium]

$1 ÷ $0.20 per kWh = 5 kWh	**[1]**

(b) Calculate how long the compressor can run on $1 of electricity. [2 marks · Hard]

5 kWh = 5,000 Wh	**[1]**
Energy = power × time, so time = 5,000 Wh ÷ 1,250 W = 4 h	**[1]**

Common slip: *5 ÷ 1,250 = 0.004 h mixes kWh with W. Change kWh to Wh first.*

Section E: Mixed multi-step scenarios (19 marks)

E1 Cable car to a lookout (7 marks)

(a) Calculate the GPE the car and passengers gain on the trip up. [2 marks · Easy]

GPE = mgh = 900 × 10 × 150	**[1]**
GPE = 1,350,000 J (1,350 kJ)	**[1]**

(b) The GPE is the useful energy. Calculate the electrical energy the motor system uses for the trip. [2 marks · Hard]

Useful energy = 75% of the energy in, so energy in = 1,350,000 ÷ 0.75	**[1]**
Energy in = 1,800,000 J (1,800 kJ)	**[1]**

Marking note: *Self-check: 75% of the answer should give back the 1,350,000 J of GPE.*

(c) Calculate the energy wasted on the trip. [1 mark · Medium]

Wasted = 1,800,000 − 1,350,000 = 450,000 J	**[1]**

Marking note: *Mark consequentially: use your answer to part (b).*

(d) On the next trip, the passengers are lighter, so the car and passengers have a total mass of 600 kg. The climb is still 150 m and the motor system is still 75% efficient. Calculate the electrical energy it uses for this trip. [2 marks · Hard]

GPE = 600 × 10 × 150 = 900,000 J	**[1]**
Energy in = 900,000 ÷ 0.75 = 1,200,000 J	**[1]**

E2 Brakes on a mountain road (6 marks)

(a) Calculate the car's kinetic energy at 15 m/s. [2 marks · Easy]

KE = ½ × 1,200 × 15^2 = ½ × 1,200 × 225	**[1]**
KE = 135,000 J	**[1]**

(b) Name the force that acts where the brake pads rub on the brake disc. [1 mark · Easy]

Friction	**[1]**

(c) Assume the four brake discs share the thermal energy equally. Calculate the thermal energy each disc gains. [1 mark · Medium]

135,000 ÷ 4 = 33,750 J	**[1]**

(d) Each metal brake disc has vents: gaps that air flows through as the wheel turns. Explain how the vents help a hot disc cool down. [2 marks · Medium]

Heat passes from the hot disc into the air in the vents, and the moving air carries it away. This is convection.	**[1]**
Fresh air keeps flowing past the disc, so heat keeps moving from the metal into it.	**[1]**

Marking note: *Also accept for the second mark: the vents give the air more of the disc's surface to touch. Reject "the vents let the heat out" or "the cold air goes in" with no idea of moving air carrying heat away.*

E3 Is this claim correct? Ski lodge bunk room (6 marks)

(a) Calculate the room's heat loss per hour after both stages. [2 marks · Easy]

Total saving = 0.45 + 0.45 = 0.9 kWh per hour	**[1]**
Heat loss after = 3.0 − 0.9 = 2.1 kWh per hour	**[1]**

(b) The manager says: "Insulating the floor has cut the room's heat loss in half." Is she correct? Use a calculation to decide. [2 marks · Hard]

Not correct (a verdict backed by a number).	**[1]**
Half of 3.0 is 1.5 kWh per hour, but the loss is now 2.1 kWh per hour. The cut is 0.9 ÷ 3.0 = 0.3, so 30%, not 50%.	**[1]**

Marking note: *Reject "No, it's less than half" with no numbers.*

(c) Calculate how much less heat the room loses over a 12-hour night after both stages. [1 mark · Medium]

0.9 kWh per hour × 12 h = 10.8 kWh	**[1]**

(d) The floor insulation also comes as R4.0. State which one, R2.0 or R4.0, slows heat flow more. [1 mark · Easy]

R4.0. A higher R-value insulates better.	**[1]**

Extension (bonus marks)

X1 Medical delivery drone **Hard** (4 marks)

(a) Calculate the drone's GPE and KE when it is 90 m up and flying at 15 m/s. Add them to find its total useful energy. [2 marks]

GPE = 2 × 10 × 90 = 1,800 J	**[1]**
KE = ½ × 2 × 15² = 225 J. Total = 1,800 + 225 = 2,025 J	**[1]**

(b) Calculate the efficiency of the drone for this trip. [1 mark]

2,025 ÷ 6,750 × 100% = 30%	**[1]**

(c) Name one form the wasted energy takes. [1 mark]

Thermal energy (heat) in the motors, battery and wires, or sound from the propellers	**[1]**

X2 Solar-powered radio **Hard** (3 marks)

(a) Calculate the percentage of the sunlight's energy that reaches the radio as electrical energy. [1 mark]

Efficiencies multiply along a chain: 0.18 × 0.85 = 0.153, so 15.3%	**[1]**

(b) The rangers can afford one change: a 24% panel, or a battery that gives back 95%. Which change gives the greater overall efficiency? Show calculations. [2 marks]

New panel: 0.24 × 0.85 = 20.4%. New battery: 0.18 × 0.95 = 17.1%	**[1]**
The new panel gives the greater overall efficiency (20.4% is more than 17.1%).	**[1]**

Mark summary

Section	Skill	Questions	Marks
A	Kinetic energy	A1, A2, A3, A4	10
B	Gravitational potential energy and energy chains	B1, B2, B3	11
C	Efficiency and Sankey diagrams	C1, C2, C3, C4	15
D	Power, energy and running cost	D1, D2, D3, D4	14
E	Mixed multi-step scenarios	E1, E2, E3	19
Total			**69**
Extension (bonus)			7